

Digital In-Memory Compute for Machine Learning Applications With Input and Model Security

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Abstract—Digital in-memory compute (IMC) architectures allow for a balance of the high accuracy and precision necessary for many machine learning applications, with high data reuse and parallelism to reduce energy consumption. However, one often overlooked parameter is security, which is necessary to maintain the privacy and integrity of the accelerator. In this work, we propose an IMC macro design that is protected against two types of eavesdropping attacks, passive physical side-channels and memory bus-probing. This is achieved through secure compute that eliminates the need for random bits, local model decryption with a lightweight cipher, and secret key generation reusing existing IMC circuitry. These contributions provide side-channel security against all practical attackers beyond 1 million samples, while still operating without any effect on neural network accuracy at 8.1 TOPS/W energy efficiency.

Index Terms—In-memory compute (IMC), machine learning, physically unclonable functions, side-channel security.

I. INTRODUCTION

MACHINE learning accelerators are specialized hardware for energy-efficient implementations of neural networks required for a wide variety of applications, including natural language processing, computer vision, and speech recognition tasks. In contrast to traditional machine learning accelerators [1], in-memory compute (IMC) co-locates the memory and computation to minimize data transfer energy, while taking advantage of high parallelism and data reuse to amortize costs [2], [3], [4]. This is particularly useful since many modern applications require large amounts of data, and the energy of loading all this data far exceeds that of performing the computation. While many initial implementations used

current- or charge-based computing within memory cells to perform multiply and accumulate operations in the analog domain, more recent works have proposed digital IMC to achieve some of the benefits of analog IMC, while still providing the high bitwidths and accuracy levels necessary for many practical workloads, especially with technology scaling [5], [6], [7]. Digital IMC interleaves the memory bitcells and computation logic to minimize data transfer distance and energy but still performs the computation with logic gates that are robust to noise and mismatch.

ML workloads contain security risks that can be exploited using physical attacks [8], [9], [10] to affect the privacy and integrity of the device. The input activation to the accelerators can reflect private data that is being collected on an IoT device, such as images of faces or voice recordings, which should not be leaked to an untrusted entity. Furthermore, knowledge of the model stored in an accelerator can reveal multiple vulnerabilities. For one, the weights are generally trained using private training datasets that can be revealed once the model is known [11], [12]. In addition, knowledge of the weights of an accelerator allows an attacker to perform targeted adversarial attacks through methods such as RowHammer [13], [14], [15], which affects the computation integrity. Finally, the process of training these specialized ML algorithms generally requires a significant amount of man and machine hours and thus must be protected as private IP for competitive advantage.

This work proposes a secure digital IMC accelerator that can protect against SCAs and BPAs. The key contributions are listed below.

- 1) Computing with SCA secure MACs that take advantage of natively secure gates to eliminate fresh random bits and reduce latency, with no impact on neural network accuracy.
- 2) Performing model decryption with a lightweight cipher on-chip for BPA security.
- 3) Generating a secret key on-chip to maintain statistical security guarantees of the cipher with a minimal overhead physically unclonable function that reuses IMC SRAM.

Section II describes the threat model and assumptions we make about the attacker, as well as some prior work that addresses these same scenarios. Section III discusses the key contributions of this work. The security and performance measurement results of this implementation are provided in Section IV. Section V concludes and suggests some directions for future work.

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II. BACKGROUND

A. Threat Model

We assume that the attacker can perform two types of attacks—physical side-channel and bus probing attacks (BPAs). Other attacks, such as software-based or micro-architectural side channels, active fault injection, and foundry Trojan insertion are out of scope for this work and can be addressed with parallel protections proposed in literature [16], [17], [18], [19].

Side-channel attacks (SCAs) use unintended leakage sources to reconstruct data or operations on-chip [8], [20], [21]. Specifically, physical power and electromagnetic (EM) SCAs take advantage of the direct relation between CMOS circuit operation and the current consumption. Power SCAs measure these different currents directly by measuring the voltage drop across a shunt resistor on the power rail [22]. However, EM SCAs can non-invasively use a probe to measure the EM fields resulting from the spatially distributed current flows [23]. For this work, we assume that attackers are limited to a practical number of samples they can collect before their presence is detected. Using these side-channel traces to perform template- or correlation-based attacks, an attacker can passively predict the data being processed in the computation operation.

Bus-probing attacks are also relevant to ML accelerator applications. Since standard models are too large to be stored entirely on-chip, weights are fed from off-chip memory during the inference operation [1]. With traditional packaging methods, this happens across exposed PCB traces for data lines that can easily be tapped by the attacker to read out the model [24]. Furthermore, storage of the model in off-chip non-volatile memory allows for cold-boot attacks that can recover data even when the device is powered off [25], [26]. A BPA on a naive implementation can easily reconstruct the entire neural network structure by collecting the digital weights and biases during the data loading and computation of each layer.

These attacks can be performed with physical access to the system being attacked, or through remote or far-field attacks that have been proposed in prior work [27], [28]. We also assume that the attacker can perform full reverse engineering of the chip for complete knowledge of the architecture down to the transistor-level circuits [29]. This allows the attacker to know the control flow and macro operation, but it is still too expensive to quantify small mismatch-dependent variations in fabrication for a large number of chips.

B. Side-Channel Security

Prior works have addressed ML side-channel security through algorithmic masking and equalization methods [30], [31]. Maji et al. [30] addressed SCA security for traditional accelerators through a threshold implementation (TI), where each computation is split up into decorrelated shares. By applying this protection to all neural network computations, the accelerator is secure for an infinite number of attack samples. The area and energy overheads are lowered somewhat by using a shift operation to perform multiplies, but this poses a limitation that weights must be powers of 2. In addition, the

ideal TI design requires a constant source of random bits every clock cycle.

An empirical protection that is instead intended for practical limitations is provided in [31]. Some security is provided by shuffling the MAC operations in time and space across the processing elements. In addition, the power consumption for each set of inputs and weights is estimated and compensated for with an on-chip DAC that attempts to have constant total chip power consumption. While this method provides a voltage scaling agnostic and block-level protection, it also requires a random sequence generator for the shuffling.

Both these prior works focus on security for conventional accelerator architectures. While the requirements for constant random bits were not a limitation in these implementations, they cannot be applied to IMC. IMC macros have thousands or even millions of multiply-and-accumulates (MACs) in parallel each clock cycle. If each required even a small number of random bits, this would become prohibitively expensive with the pseudorandom number generation (PRNG) cost overtaking the computation. Thus, any security scheme for IMC should eliminate the need for fresh random bits during the operation.

C. Bus Probing Security

BPA security has been provided in prior works through block-level algorithmic or fine-grained circuit-level protections. Maji et al. [30] and Fang et al. [31] used a trusted execution environment-like architecture, where the model is stored in an encrypted form off-chip, and only decrypted on-chip when needed. By using ciphers such as AES [31] and Trivium [30], these works provide cryptographic security guarantees. However, this mathematical security assumes the presence of a secret key. If this key must be read from off-chip, that transmission can be probed and the security of the cipher is lost.

Security from probing attacks of non-volatile IMC data is instead focused on in [32]. Each bitcell stores the correct value and its inverse, and decryption is performed as part of the compute by charging or discharging the bitline. This implements a low overhead XOR cipher but can lead to security flaws if the key is reused. Thus, we need a key as long as the entire neural network model [33], which would be expensive to generate and provide from off-chip.

III. CHIP DESIGN

The overall architecture of the macro with key contributions highlighted is shown in Fig. 1. The baseline architecture follows that of [5], using a bit-serial computation of 1–8 bit activations and weight precisions that are a multiple of 4. The SRAM consists of eight columns, each with 128 rows of 4-bit weights. The columns are interleaved with 1-bit XNOR multiply gates, and each column contains an adder tree and accumulator. Multiple columns can be concatenated for higher weight precision. In addition, low area and power overhead model decryption and key generation circuits are included external to the SRAM bank.

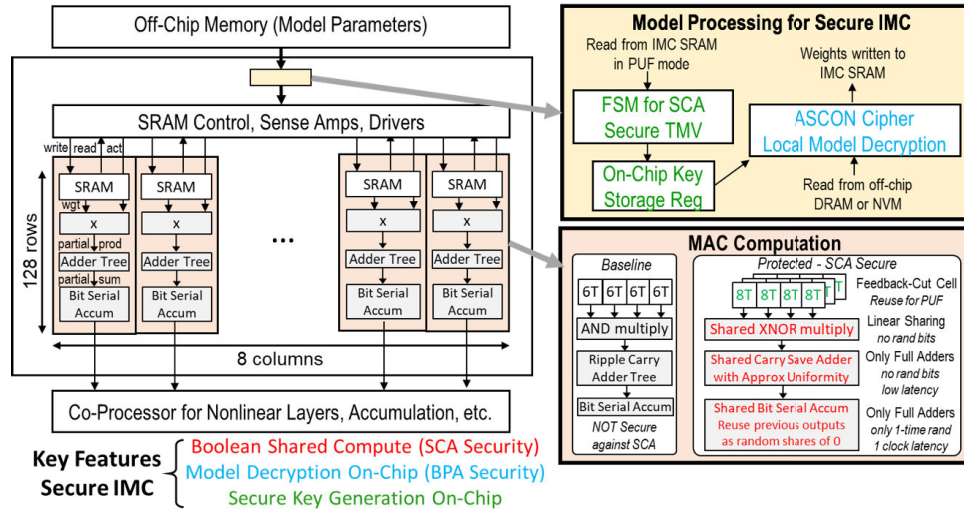


Fig. 1. Block diagram of the proposed secure macro with Boolean-shared digital IMC for SCA security, model decryption on-chip for BPA security, and a secure key generated on-chip.

A. Side-Channel Secure Compute

The MAC computation is done using a Boolean-sharing methodology for side-channel security. The secret sharing scheme is similar to cryptographic multi-party computation, where n parties split information in a way that it can only be recovered if at least x of them collude [34], [35]. With Boolean masking, every digital bit is split such that all n shares (which are seemingly random when observed alone) are required to know the original value [36], [37], [38]. Each logic function is also split to receive and operate on strict subsets of the input shares, such that each subfunction's computation and thus the total circuit's power consumption is unrelated to actual data.

In this work, a Boolean-sharing implementation with three shares is chosen to achieve first-order security. As in prior work for neural network side-channel protections in realistic conditions [30], [39], [40], we assume that the eavesdropper is limited to first-order attacks and cannot collect enough data for higher-order attacks [41], [42], [43]. This is especially fitting in IMC, with the high number of bits being processed in parallel. If necessary, the design described here could be directly extended with more shares and more complex shared functions for higher-order security.

One particular type of Boolean sharing, called a TI, provides provable security against SCA attacks of various orders, but it comes at the cost of large overheads [30], [37]. Specifically, one of the most difficult conditions to satisfy is uniformity, which requires that each of the shared representations has equal probability for any given unshared input. This ensures that only bias in the distribution of unshared values contributes to any bias in the shared version [44]. Some common functions, such as AND gates, do not produce uniform outputs given uniform inputs with a minimal number of shares. This can be solved by using more shares, at the expense of much larger circuits [45]. Alternately, the output shares can be refreshed with independent random bits to generate uniform inputs for the successive gates [46].

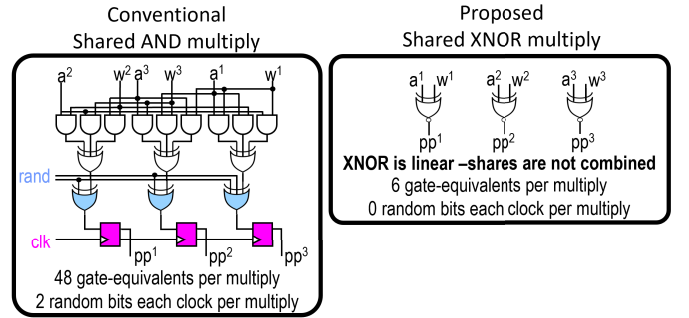


Fig. 2. Comparison of conventional AND and proposed XNOR multiply in shared format.

In this work, we use a relaxed uniformity requirement that targets practical attacker levels rather than ideal security and saves on these overheads. Specifically, we require that our circuits satisfy the following properties for a function $t = F(b, c)$ shared as $(t^1, t^2, t^3) = F^{sh}((b^1, b^2, b^3), (c^1, c^2, c^3))$.

- 1) *Correctness*: If $b = b^1 \oplus b^2 \oplus b^3$, $c = c^1 \oplus c^2 \oplus c^3$, then $t = t^1 \oplus t^2 \oplus t^3$.
- 2) *Non-Completeness*: If $F^{sh} = \{F^1, F^2, F^3\}$, then each of F^j uses a proper subset of $\{b^1, b^2, b^3\}$ and $\{c^1, c^2, c^3\}$.
- 3) *Approximate Uniformity*: Each individual sub-circuit output is uniform. Non-jointly uniform outputs are combined after approximate refreshing (not with random bits).

Approximate uniformity is particularly acceptable for IMC implementations, which have many parallel operations with unrelated weights and activations, and thus higher attacker SNR compared to typical accelerators.

1) *Multiplier*: In a typical bit-serial computation with multi-bit weights and activations [5], multiply operations are implemented with AND or NOR gates. However, these contain nonlinear logic, so the multiply between each activation and weight share cannot be performed independently. Furthermore, these require remasking with random bits or utilizing more input shares to maintain uniformity and registers on the outputs to maintain non-completeness and glitch resistance [43].

Implemented in standard TI form, each AND 1-bit multiplier would require the logic shown in Fig. 2, which contains 48 gate equivalents and 2 random bits per clock cycle. However, this is not scalable with in-memory compute, where there are many parallel multiply operations.

We can achieve significant savings by using a natively secure linear XNOR gate for the multiply. The linearity allows for computing each share independently and maintains uniformity and glitch resistance without random bits or registers after the multiply. The standard four transistor transmission gate implementation can be used for the XOR gates, for a total of six gate equivalents per multiply (each XNOR has four additional transistors for inverters on the weight inputs to balance the SRAM bitcell loads for the PUF operation described in Section III-B2) and no random bits necessary. However, directly substituting AND for XNOR gates will not lead to a proper computation. For arithmetic correctness, some modifications to the neural network compute are necessary.

In general, XNOR-based multiplication is used for binary neural networks, where working with +1 and -1 values provides better range and efficiency compared to 0 and 1 [47], [48]. However, most practical tasks require higher bit precisions. Traditionally, since both activations and weights need to be in the same format, implementations with binary weights and higher precision activations require both to be in 2's complement representation. Instead, [7] stores the multibit activation as a sum of powers of 2 multiplied by +1 and -1, so that binary weights can also be +1 or -1, which restores neural network accuracy.

A similar idea is utilized in this work to maximize the usage of natively secure XNOR gates, but without limiting the weights to binary. We propose a method to use multi-bit XNOR computation for computing the MAC of multi-bit weights and activations to reduce Boolean-sharing overheads. The details of this method are presented below.

We first describe the case of signed weights and signed activation values. Each signed value (SV) is a sum of binary weighted +1 and -1 s, $SV = \sum_{i=0}^n 2^i \times (-1)^{1-b_i}$. This is represented in a binary value (BV) format on which computation is performed, which ranges from all 0 s (minimum negative) to all 1 s (maximum positive). While the SV is limited to odd values, we can store a $2 \times$ wider range in values (e.g., 4 bits can store -15 to 15), so the neural network can be retrained without any expected loss in accuracy. The conversion between SV and BV can be expressed as $SV = 2BV - (2^n - 1)$.

We now prove that multiplication with XNOR and standard binary addition (output of IMC is BV_{out}), as defined in (1), provides the correct MAC value after constant factors, as shown in (8) are applied (output after conversion is SV_{out}). For this, we define the bitwidth of the activation as n_a , the bitwidth of the weight as n_w , and the number of inputs (IMC rows) being added as r . The IMC computation works in a bit serial activation format [49], [50]

$$BV_{out} = \sum_{i=n_a-1}^0 2^i \sum_{j=0}^r \sum_{k=0}^{n_w-1} 2^k \times \overline{b_{a,ij} \oplus b_{w,jk}} \quad (1)$$

$$BV_{out} = \sum_{i=n_a-1}^0 2^i \sum_{j=0}^r \begin{cases} \sum_{k=0}^{n_w-1} 2^k \times b_{w,jk}, & \text{if } b_{a,ij} = 1 \\ \sum_{k=0}^{n_w-1} 2^k \times \overline{b_{w,jk}}, & \text{if } b_{a,ij} = 0 \end{cases} \quad (2)$$

$$BV_{out} = \sum_{i=n_a-1}^0 2^i \sum_{j=0}^r \begin{cases} BV_{w,j}, & \text{if } b_{a,ij} = 1 \\ \overline{BV_{w,j}}, & \text{if } b_{a,ij} = 0. \end{cases} \quad (3)$$

By the definition of the SV and BV conversion, we find that inverting BV is the same as negating SV. Thus,

$$BV_{out} = \sum_{i=n_a-1}^0 2^i \sum_{j=0}^r \left(\frac{(-1)^{1-b_{a,ij}} SV_{w,j} + 2^{n_w} - 1}{2} \right) \quad (4)$$

$$BV_{out} = \sum_{j=0}^r \frac{SV_{w,j}}{2} \sum_{i=n_a-1}^0 2^i (-1)^{1-b_{a,ij}} + \sum_{j=0}^r \sum_{i=n_a-1}^0 \frac{2^i (2^{n_w} - 1)}{2} \quad (5)$$

$$BV_{out} = \sum_{j=0}^r \frac{SV_{w,j} \times SV_{a,j}}{2} + \frac{2^{n_w} - 1}{2} \times r \sum_{i=n_a-1}^0 2^i \quad (6)$$

$$BV_{out} = \frac{SV_{out}}{2} + \frac{(2^{n_w} - 1)(2^{n_a} - 1)}{2} \times r \quad (7)$$

$$SV_{out} = 2 \times BV_{out} - (2^{n_w} - 1)(2^{n_a} - 1)r. \quad (8)$$

This expression provides the correction factor for the true accumulator output from the one computed by the macro, with constant terms determined by only the network bitwidths and sizes, which can easily be precomputed off-chip.

A similar method can also be used for unsigned activations with signed weights, such as those that result from the output of ReLU activation functions. In this case, the unsigned value (UV) of the activation and BV used for computing are the same. We can find the correction factor similarly for XOR multiplication, by continuing from (6), to get $SV_{out} = BV_{out} - (((2^{n_w} - 1)(2^{n_a} - 1)r)/2) + ((2^{n_a} - 1)/2) \sum_{j=0}^r SV_{w,j}$. While this factor does require the sum of all weights in the MAC operation, this does not change for different input activations and can also be precomputed and stored.

We also confirm that restricting weights and activations to odd values only has no effect on neural network accuracy with some test cases shown in Table I. Quantization-aware training [51] is used for both standard and odd quantization results. Since odd quantization does not limit the amount of information that can be stored in the network, and the digital IMC computation is only affected by quantization noise, we expect these results to apply to other test cases and networks without loss of generality.

2) *Adder Tree*: After performing the multiplication with XNOR gates, the partial products from the rows are combined with standard digital addition. In traditional digital IMC implementations, this is done with a ripple carry adder tree [5]. While the sharing of the carry computation for a full adder is natively uniform, that of a half adder is not since it uses an AND gate. Thus, we avoid half adders to eliminate the need for random bits for refreshing. In addition, even though the sum computation is linear with shares computed independently,

TABLE I

ACCURACY COMPARISON OF STANDARD QUANTIZATION (ODD AND EVEN VALUES) AND XNOR-MULTIPLY QUANTIZATION (ONLY ODD VALUES)

Test	Original (fp)	Standard Quant	XNOR Quant
AlexNet CIFAR-100	59.2% top-1	60.1% top-1	60.0% top-1
4b-weight, 8b-act	84.5% top-5	85.0% top-5	84.7% top-5
AlexNet ImageNet	56.5% top-1	49.7% top-1	50.2% top-1
8b-weight, 8b-act	79.1% top-1	74.6% top-5	74.8% top-5
ResNet18 ImageNet	69.8% top-1	63.7% top-1	63.8% top-1
8b-weight, 8b-act	89.1% top-5	85.6% top-5	85.5% top-5
MobileNetv2 ImageNet	71.9% top-1	62.1% top-1	62.1 % top-1
8b-weight, 8b-act	90.3% top-5	84.5% top-5	84.3% top-5

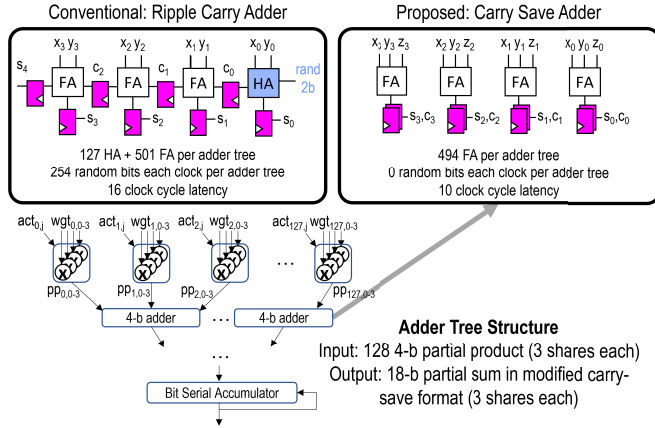


Fig. 3. Benefit of carry-save adder compared to ripple-carry for shared implementations, as well as structure of adder tree in MAC computation logic.

a register is required on each output to maintain timing synchronization with the carries (which require registers for non-completeness).

However, latency can be decreased significantly by using a carry-save adder format, where carry bits are not propagated within the same add, as shown in Fig. 3 compared to the ripple carry adder. Furthermore, by combining three operands at a time, no half-adders are necessary in the adder tree. For a standard carry save adder tree, the output of the last stage is in carry save format, where each bit position has 2 bits of information per share (carry and save values). We reduce the latency further by eliminating the last two pipeline stages of the adder tree and leaving the partial sum in a modified carry-save format where some positions have 1 or 3 bits of information instead. These last few additions are then performed along with the bit-serial accumulator computations.

As mentioned in [52], the sum and carry output of each adder are not jointly uniform. However, these values are never combined directly, as they refer to different bit positions. Instead, at least one of these outputs will pass through another adder stage before being recombined, regardless of the specifics of the digital logic synthesis. This is not always enough to recover perfect joint uniformity of the values. However, this intermediate logic with other unrelated bits significantly reduces the biases in the sharing so that it is approximately uniform. For the limited number of traces in practical attacks, this slight deviation from ideal uniformity is acceptable and will not lead to leakage of information.

For example, we look at three cases of a full adder with different inputs: 1) when carry and sum are combined directly;

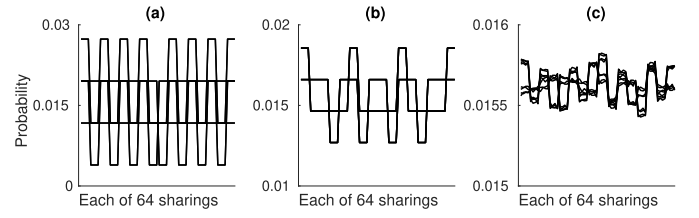


Fig. 4. Plot of the probabilities of the possible sharings of each unshared set of inputs to the full adder, along with the average coefficient of variation. (a) CV = 0.44. (b) CV = 0.11. (c) CV = 0.01.

2) after the sum output is passed through an unrelated adder stage; and 3) after both the sum and carry outputs are passed through unrelated adder stages. We depict the uniformity of each of these options in Fig. 4, which shows the probability of the 64 $((2^2)^3)$ different sharings for each of 8 (2^3) possible triplets of unshared inputs to the full adder. As we can see, the standard deviation becomes negligible compared to the mean when we combine the sum and carry more indirectly, implying approximate joint uniformity.

3) *Bit Serial Accumulator*: The adder described in Section III-A2 generates a partial sum. This is then combined over multiple clock cycles bit serially to multiply the multi-bit activation by the weights. To gain the same benefits of low latency and no random bits for half adder refreshing, accumulation is also performed with a carry-save adder with a modified carry-save adder representation. The reduction of bits is done in two stages (completed in 1 clock cycle on positive and negative clock edges).

However, when computing the first accumulation operation (MSB of activation), some of the inputs of the previous accumulation value need to be reset to zero. Adding shared secret values with known zeroes is not secure, so it is necessary to modify the compute operations to account for this, with the options shown in Fig. 5. When possible, this can be done by substituting addition operations with direct assignment of the secret values.

However, this is not possible for 7 of the full adders that contain reset 0 values. Instead of requiring 14 actual random bits from a constantly running PRNG to represent the zeros in a shared form ($s^1 = r_1$, $s^2 = r_2$, and $s^3 = r_1 \oplus r_2$), we can use a method inspired by [53]. Here, we use 2 shares of accumulator bits from previous computations to create random shares of zero. Thus, fresh random bits are only required one-time at the start of the operation for the chip. The main requirement for these random shares of zero is that these bits are jointly uniform with the corresponding partial sum bits, even when the shared weight bits are the same between computations. As with the design of the adder, we only require computational rather than mathematical uniformity to reduce overheads of remasking. This is confirmed with functional Monte Carlo functional simulations of the compute circuitry for different biases in the activation and weight input values, as well as different frequencies of weight update.

B. BPA Security

All neural network computations require weight and activation inputs. For IoT applications where attackers have easy access to devices, we assume the activations are fed from

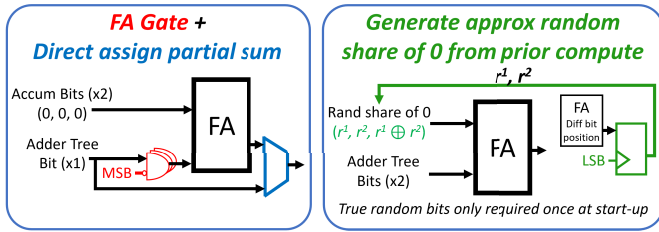


Fig. 5. Special methods during MSB accumulation computation to maintain uniformity.

an SCA secure ADC such as [54], [55], and [56]. However, for practical neural networks, there are too many weights to store completely on-chip. Thus, these must be fed from an off-chip DRAM or NVM. Any SCA protections on-chip are irrelevant if the attacker can directly measure the buses transferring weight data between the DRAM and custom IC, or perform a cold boot attack on the NVM. For this reason, memory encryption should be used for off-chip storage [57]. The weights can then be decrypted on-chip when needed before being used in compute.

1) *Lightweight Cipher*: A cryptographic cipher for encrypting and decrypting the weights provides mathematical guarantees of security, but we do not want this component to add a large area or power overhead. In addition, the cipher itself must be implemented in an SCA-secure fashion to avoid information leakage during the decryption. We use the ASCON cipher [58], the winner of the NIST Lightweight Cryptography competition [59], which has been well-proven for its area and energy efficiency. Key-generation and sharing can be coordinated in advance through the PUF described in Section III-B2. In addition, the cipher is relatively resilient to SCAs, since determining the cipher's state does not directly lead to recovering the secret key for breaking future decryption operations [58]. Furthermore, the ASCON implementation is conducive to masking, with the lowest percent of AND gates (which have high Boolean-sharing overheads) among lightweight ciphers [60]. The other operations in the cipher (NOT, XOR, and rotate) are all linear operations that can be performed independently on the shares with minimal overhead.

Daemen [53] proposed a “Changing of the Guards” implementation of the Keccak [61] permutation, which uses various bits of the state to remask other bits to maintain uniformity in the nonlinear permutation. This method only requires 4 external random bits at the start of the operation and no constant random bits during the operation of the cipher. Since the base nonlinear step of both the Keccak and ASCON permutation are the same, we can use this same method for low overhead threshold implemented SCA secure ASCON [62]. The block diagram of the model decryption process is shown in Fig. 6.

2) *Secret Key Generation*: Cryptographic ciphers to decrypt weights on-chip require a secret key, but feeding the key value from off-chip is inherently insecure and can nullify the security of later operations of the cipher. Techniques such as e-fuses are vulnerable to SEM imaging that can read out whether each fuse is connected [63]. Instead, physically unclonable functions (PUFs) can be used to securely generate keys [64], [65]. During manufacturer testing, a table of challenge and response pairs of the PUF can be collected for each chip. Later

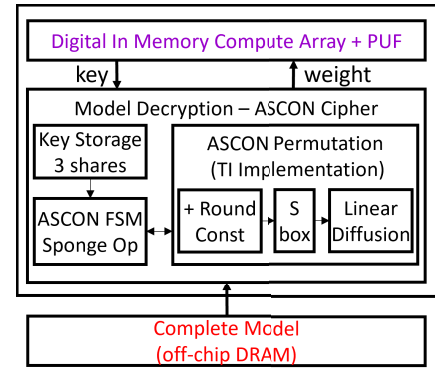


Fig. 6. Block diagram of ASCON cipher implementation.

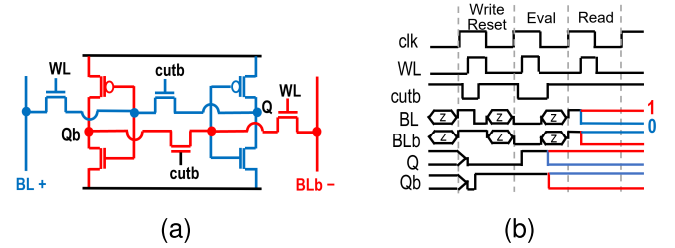


Fig. 7. SRAM (a) bitcell and (b) timing diagram for use as PUF. If the right (left) pull-down dominates, the PUF generates a 0 (1).

upon deployment, a certain key can be used by the cipher simply by evaluating the PUF on that specific challenge, without being measurable to an external observer. The security of the ASCON cipher allows for 2^{128} decryptions without changing the key, which is a high enough security level for any practical usage. In addition, the key can be refreshed occasionally with a new challenge to achieve additional security.

Even so, a large PUF bank can be a significant area overhead on the IC. Instead, we propose reusing the IMC circuitry for the PUF. Much of the IMC consists of SRAM cells, which are conducive to PUF circuits. Prior SRAM-based PUFs require overwriting entire banks or multiple rows of SRAM to get responses from one address [66], [67], [68]. Others require additional peripheral circuitry such as level shifters and time to digital converters to generate the PUF bits [69], [70]. Here, we propose a SRAM PUF that requires only a small additional block of synthesized digital logic, by using the SCA secure SRAM [Fig. 7(a) and (b)] of [71] as the bitcell. The bitcell has a standard cell-type layout, with each set of three vertically stacked cells forming one set of Boolean shares. This arrangement leads to three differential bitlines vertically routed within the width of each bitcell, where each pair is routed on a different metal layer to reduce coupling between shares without increasing the bitcell size further. Each bitcell is also adjacent to the custom XNOR multiply gate, which has horizontal connections to the activation driver and local internal bitcell nodes.

An effective SRAM PUF requires a method of overriding the stored data with the intrinsic preferred bitcell state due to manufacturing variation (bits of key). The side-channel secure bitcell already contains feedback-cut transistors that can be used for this purpose. First, the feedback is cut and the inputs to both separated inverters are precharged. After this, the feedback can be re-connected so that the SRAM settles to

0 or 1 based on which inverter is stronger, which can be read out with standard SRAM read. Only the pre-existent data at a specific wordline is erased as part of this evaluation, and this can be minimized through sub-divided wordlines. This leads to a noise-resistant and low-overhead PUF that can be used throughout chip operation to generate reliable keys on demand. Furthermore, error-correction and entropy extraction can be added to further improve the randomness and robustness [72].

Resilience to noise is added through temporal majority voting (TMV) [73] with successive evaluations of the same bitcell. We allow a configurable number of evaluations to set the robustness per use case. Since the finite-state machine (FSM) is working on small groups of four key bits, it does leak some data statistics even with Boolean sharing, particularly since the repeated TMV operations increase attacker SNR. For example, if we see power spikes corresponding to 11 counter increments, it is known that three unshared bits are 1 s, and the other is 0. Though this does not lead to full recovery of the key, it may provide valuable information that reduces the difficulty of brute force attacks on the cipher.

For this reason, we protect against this simple power analysis attack with differential operation in the TMV logic. The sense amplifier already provides differential values after the read, and both of these are provided to the FSM. This generates constant switching activity regardless of the specifics of the data and is enough to secure the key generation from SCAs combined with the existing Boolean sharing.

We expect some degradation in PUF performance due to small systematic differences being amplified. Biases in PUF bits can be fixed through trimming, where cells with strong dependence on external factors and less intrinsic mismatch can be masked [73], [74], [75] or have debiasing methods applied [76], [77]. While negative-bias temperature instability aging typically affects SRAM PUFs, the near randomness of the weights stored during normal operation alternates which transistor is experiencing aging and partially offsets the effects [78]. Furthermore, noise resilience can be improved through burn-in hardening. This method has been used in similar cross-coupled bitcells to direct aging for each cell in the direction of the existing mismatch, forcing it to have an even greater propensity for settling in its preferred direction [73].

IV. MEASUREMENT RESULTS

This design, shown in Fig. 8(a), was taped out in 14-nm CMOS technology. Multiple facets of the chip are evaluated as detailed in Sections IV-A–IV-C.

A. Side-Channel Security

The independent digital circuits are individually implemented on a Chip Whisperer [79] CW305 FPGA platform [Fig. 8(b)] to achieve high targeted SNR and to demonstrate the strong SCA security guarantees in real-world scenarios. For example, the independent pipeline stages (and even sub-parts of just one) can be implemented and attacked separately to minimize background switching. If SCA security remains beyond 1 million measurements in even these ideal

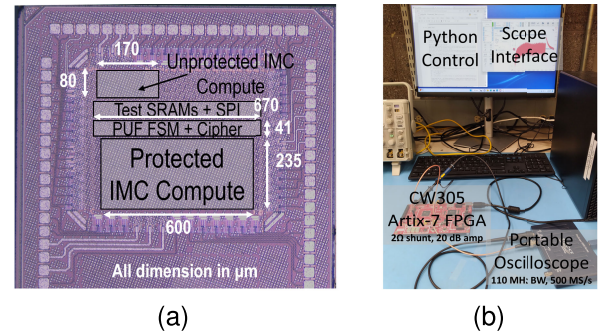


Fig. 8. (a) Custom ASIC micrograph and (b) Chip Whisperer targeted SNR setup.

tests, this implies even stronger security guarantees in real-world scenarios, once parallel operations and unrelated activity lower the attacker SNR. Due to the algorithmic nature of Boolean sharing protections, security to SCA on an FPGA also holds for custom IC implementations. Power SCA security directly translates to EM SCA security, since computations of individual shares will be too spatially close to resolve with current EM probes. Furthermore, testing on a dedicated attack platform like the Chip Whisperer allows for artificially boosting attacker SNR with multiple parallel copies and utilizing the proven high SNR measurement circuitry.

Correlation power analysis (CPA) simulates a realistic attack on the system where the attacker controls one varying input and tries to recover a fixed secret input [80]. For compute logic, we consider the varying input to be activations, and the fixed input to be model weights. Since weights are stationary in the IMC SRAM throughout the computation, we maintain the same sharing of the weights throughout all of the attacker measurements. However, since activations are streaming in real-time, these are newly shared for each sample. The results for CPA of each sub-block of the SCA-protected multiply and adder tree are shown in Fig. 9(a), when compared to the unprotected (non-pipelined) version in Fig. 10(a). We see that the correct secret value (red) has indistinguishable correlation values from the other incorrect secret value guesses (gray) for each individual compute block for the protected implementation. The same is true for each of the two accumulator pipeline stages shown in Fig. 9(b), when compared to the unprotected accumulator in Fig. 10(b).

We also perform CPA for the cipher in the Chip Whisperer platform, with a fixed key and varying nonce. As is standard in the literature, SCA analysis is performed at the end of the initialization stage when the key is directly combined with the state [62], [81]. We see in Fig. 9(c) that the true key value is indistinguishable from other guesses in terms of the correlation to the power measurements.

Differential power analysis (DPA) is used to test the algorithmic security guarantees, by attempting to differentiate between a circuit operating on fixed and random secret inputs [22], [82]. By testing if there is a statistically significant difference between side channels from random weights and inputs versus a fixed weight and input, we determine whether there is any exploitable data-dependent leakage. In the high SNR attack scenario, we find that the unprotected circuits have significant differences (large T -value) with less than

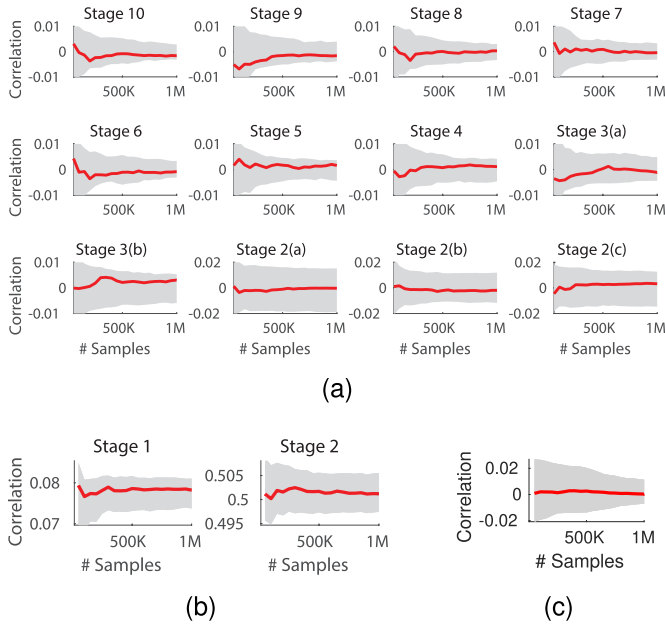


Fig. 9. CPA results—protected (a) multiply + adder tree pipeline stage sub-blocks, (b) accumulator pipeline stage sub-blocks, and (c) ASCON cipher.

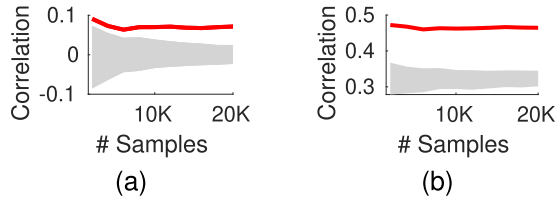


Fig. 10. CPA results—unprotected (a) multiply + adder tree and (b) accumulator.

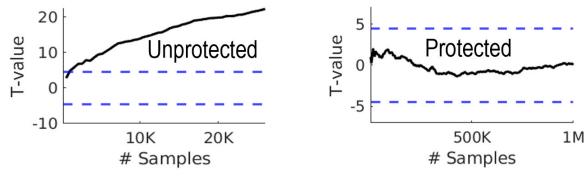


Fig. 11. DPA results on unprotected and protected ASIC compute.

1000 samples, whereas the protected circuits do not have significant differences (small T -value) even after 1 million samples.

In addition to high-attack SNR tests, we confirm the SCA security of the custom IMC on-chip with DPA. The results for the unprotected baseline and protected compute are shown in Fig. 11. As we can see, the unprotected version has large T -values with less than 5000 samples, indicating a statistically significant difference between fixed and random data. However, the protected implementation has small T -values for more than 1 million samples, so there is no reason to believe exploitable differences in side-channel traces exist.

We read from the IMC SRAM using a differential and shared TMV FSM. We confirm that this operation is SCA-resilient on the Chip Whisperer by attempting to reconstruct the key using a convolutional neural network [83], which was shown to have high SCA accuracy in prior work [54], [55]. Since we are working with Boolean-shared keys with 3 shares, we consider the rms error of guessing the unshared and shared key values. The results (Table II) show that the difficulty of

TABLE II
RMSE OF KEY-VALUE PREDICTION FROM TMV SCA

	Not Shared	Shared	Shared & Differential
Num Training Samples	250K	1M	1M
Unshared HW RMSE	1.38	4.90	7.20
Shared HW RMSE	N/A	2.35	32.13

TABLE III
PUF NIST TEST RESULTS

Test	Average P-Value
Frequency	0.703
Block Frequency	0.644
Cumulative Sums	0.776
Runs	0.345
Longest Run	0.476
Rank	0.464
FFT	0.606
Non Overlapping Template, $m = 6$ min-value across templates	0.040
Overlapping Template	0.420
Approximate Entropy, $m = 6$	0.626
Serial, $m = 9$	0.453

TABLE IV
PUF STABILITY RESULTS

Stability Mechanism	BER
Native	13.5%
TMV-7	6.93%
TMV-15	4.95%
TMV-31	3.78%
BCH Code (16383, 13856, 184)	0%

guessing the key increases significantly with Boolean-shared and differential logic protections applied, as expected.

B. PUF Performance

We collect the PUF challenge–response pairs after on-chip TMV and off-chip Von Neumann debiasing [84]. We test these against the NIST 800-22 randomness standards [85], and as in prior work with limited bits [70], we confirm that all applicable NIST tests pass for each of the five chips with all p -values larger than 0.01, as shown in Table III. Thus, we have no reason to believe the PUF sequences are not random. The most common value estimator also indicates the min-entropy for each chip is ~ 0.91 – 0.93 .

We also test the PUF's resilience to noise for various levels of TMV and error correction (Table IV). While raising the TMV time can increase stability significantly, it is not enough for the cells with very low mismatch that are instead decided by transient noise. For these, we can achieve a 0% bit-error rate with an off-chip BCH code [86]. Masking out unstable cells would also be a solution to reduce the number of error correction bits. Finally, we evaluate the inter- and intra-key hamming distance (HD) to validate the uniqueness and stability of the keys. Fig. 12 shows the intra-key HD is close to 0 (stable), whereas the inter-key HD is close to half the key bitwidth, 192 (unique).

While this initial demonstration shows the potential of the feedback-cut PUF, techniques used in orthogonal PUF works to improve performance can also be applied to this topology.

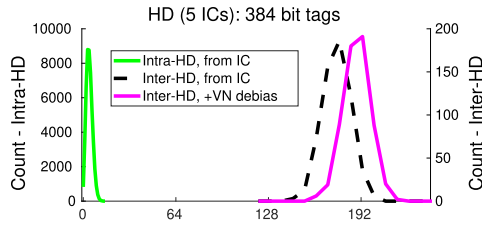


Fig. 12. HD between measurements of same key and different keys.

TABLE V

MACRO PERFORMANCE METRICS COMPARISON

Metric	Precision	Unprotected	Protected
Throughput (GOPS)	4b wgt, 1b act	41.0	81.9
	4b wgt, 8b act	9.10	10.2
Energy Efficiency (TOPS/W)	4b wgt, 1b act	90.2	6.94
	4b wgt, 8b act	14.4	0.89
Area Efficiency (TOPS/mm ²)	4b wgt, 1b act	3.01	0.49
	4b wgt, 8b act	0.67	0.061

For example, dark bit masking can turn off unstable bits to reduce TMV length and BCH code size, burn-in hardening can increase the mismatch between inverter arms and improve stability, and randomness extraction can maximize the entropy per bit [73]. In addition, the symmetry of the bitcell can be improved using careful transistor and interconnect layout for reduction of systematic variation.

C. Overall Performance

We evaluate the performance of the macro according to standard ML accelerator metrics. The throughput, energy efficiency, and area efficiency for different data bitwidths are shown in Table V for the unprotected and protected macros at a 0.55-V supply and 80-MHz operation frequency. The protected implementation achieves a maximum energy efficiency of 8.1 TOPS/W at a 0.50-V supply voltage. The scaling of the frequency is limited to 100 MHz due to the setup of the test environment, with synchronous communication and data input from the FPGA.

The throughput is higher for the protected implementation due to the pipelining of the adder tree and accumulation required for satisfying the non-completeness property of the Boolean sharing. If we also pipeline the unprotected implementation, we would see similar improvements in throughput, potentially at the expense of minor increases in area due to the additional flip-flops. While the latency of the protected version will have some overheads due to the model decryption before writing into SRAM, the computed latency dominates for typical models, so the $\sim 3.2\times$ increase in weight loading latency is not significant overall. Particularly, weight decryption operations occur quite infrequently for IMC, since the weight is loaded once and many computations are performed on it before reloading to maximize data reuse and energy efficiency, and thus any overheads to the power consumption of data loading are also amortized.

On the other hand, there are reductions in the energy and area efficiency due to Boolean sharing. The custom memory bitcell and multiply columns increase in area by a factor $5.3\times$, due to a combination of each cell being split into three

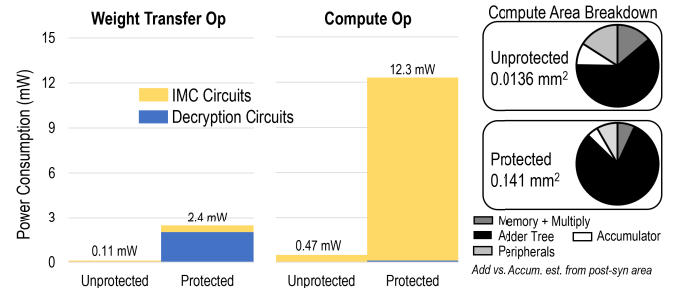


Fig. 13. Breakdown of power and area consumption of various components of secure IMC circuitry compared to baseline.

independent shares and the 2 additional PUF feedback-cut transistors. However, the major contributor to energy increase is the adder tree, which is generally the dominant component of any digital IMC implementation. The post-synthesis area of a full adder increases by $14.2\times$, which causes the entire adder tree to increase by $15.5\times$. In terms of area efficiency, this causes the adder and accumulator to be $6.4\times$ less area-efficient. The power breakdown for the various portions of the circuitry for the unprotected and protected implementation is also shown in Fig. 13. These results are in line with prior work [30], which saw a $6.1\times$ and $5.7\times$ decrease in area efficiency and energy efficiency, respectively, for the multiplier circuitry alone. While this leads to a high energy and area overhead, security cannot be achieved without some costs and our optimizations ensure that we do not have even higher overheads.

This work is compared with prior secure ML accelerators in Table VI. Comparison can be made across different architectures and applications by assuming full utilization of the macro and finding the overhead compared to its own baseline implementation. The area and power overheads include the IMC macro, cipher, and PUF FSM and are calculated as the ratio of the unprotected macro's efficiency (in TOPS/W and TOPS/mm²) to that of the protected macro. Compared to the other work, this is the first to consider digital IMC security. In addition, it includes much more rigorous testing of SCA security compared to prior IMC security work, using CPA [80] and test vector leakage assessment (TVLA) [82] tests. Furthermore, the security is added without placing additional limitations on bit precision or neural network accuracy. Most importantly, it does not require random bits during the operation and only needs one-time bits during chip configuration. With these protections, we have BPA and SCA beyond 1 million attack measurements even in ideally high-attack SNR conditions, which ensures security will last significantly longer for realistic operation.

D. Scalability

While this work presents a small macro for demonstration of the security concepts, a commercial-scale accelerator will have many macros in parallel. This can be considered similar to the case of parallel processing in chiplets, each of which performs a small part of a larger task before the outputs are integrated [87], [88]. While the individual macros will perform side-channel secure computation in the same way,

TABLE VI
COMPARISON TO PRIOR WORK

	TVLSI'21 [33]	JETC'22 [39]	JSSC'22 [30]	ISSCC'22 [32]	VLSI'23 [31]	This Work
Process	28nm NeuroSim	FPGA	28nm	22nm	40nm	14nm
Operation	-	24 MHz	0.60-0.95V 10-125 MHz	0.65-0.8V	0.70-0.90V 0.05-50 MHz	0.50-0.80V 57-100 MHz
Architecture	Analog IMC	Non-IMC	Non-IMC	Analog IMC	Non-IMC	Digital IMC
Precisions	8/16b act 8/16b wgt	1b act 1b wgt	8b act; 8b wgt (pwr of 2 only)	1-8b act 2/4/8b wgt	multi-bit	1-8b act 4/8/12/16b wgt
Threat Model	BPA	SCA	SCA,BPA	BPA	SCA,BPA	SCA,BPA
Protection	XOR Weight Encryption	Boolean Mask Shuffling	Boolean Mask Trivium	XOR Weight Encryption	Shuffling Power Comp. + AES	Boolean Mask ASCON
Random Bits	N/A	Each clk (Trivium)	Each clk (Trivium)	N/A	Each clk (RSG)	1-time Only
Overhead $TOPS/mm^2$	1.02x	4.85x LUT	2.30x	Not Reported	2.27x	6.2x
Overhead $TOPS/W$	0.71x	-	5.48x	Not Reported	1.76x	13.0x (1b act)
Security Level	One-Time Pad	TVLA ($> 2M$)	CPA, TVLA ($>> 2M$)	One-Time Pad	CPA ($> 200M$) TVLA (55M)	CPA, TVLA ($>> 1M$)

they will create macro outputs that will need to be accumulated. We expect that the side-channel secure co-processor, which combines the carry-save outputs and the shares, will also be used to combine the outputs of the separate secure macros. With many macros operating in parallel, we also have higher attacker difficulty in determining data from total power consumption, and thus the security level can be lowered to take advantage of this with less secure gates.

As the system size increases and we have multiple macros in parallel, there are a few options to ensure the weight decryption and loading overheads do not dominate. For one, we can use the ASCON-128a scheme to produce 128-bit data blocks within 8 clock cycles, rather than 64 bits within 6 clocks. Furthermore, in the case of high-bandwidth memory interfaces for loading data from off-chip, we can have a separate ASCON instance for each instance of the macro, since this only contributes 16% of the total macro area and will limit any latency overheads from decryption to our demonstrated levels. Furthermore, as the memory size increases, there will be more PUF bits available. Depending on the off-chip memory encryption procedure, either one key (generated by a combination of PUF bits across the different macros) can be used to decrypt the weights written to all macros, or a separate key can be generated and used for each individual ASCON instance. In either case, the greater availability of PUF bits ensures that the most optimal cells can be chosen through trimming or masking for key randomness and stability.

V. CONCLUSION

This work provides a generalized solution for secure digital IMC that can provide privacy and integrity in a wide variety of ML applications such as AR/VR and health monitoring. This is done by eliminating fresh random bits for security, minimizing compute latency overheads, and avoiding restrictions on neural network architectures.

Future work should focus on further decreasing the area and energy overheads of the protections, for more convenient adoption in real systems. This can be done by better characterizing the attacker's attack abilities, and potentially finding a better optimal point in the tradeoff between security level and Boolean-sharing overheads. In addition, methods such as approximate computing [7] can be used for further exploitation of natively secure compute gates, which still allow for high accuracy due to the resilience of neural network computations.

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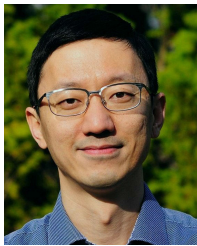
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